Same Facts, Different Interfaces: Evaluating FAQ, Retrieval Chatbot, and AI Chatbot for Student Services

Henry Kofi Dodzi Dakpo

Abstract

Universities increasingly use digital interfaces to provide student-service information, yet their user experiences may differ. This study compared an FAQ webpage, retrieval chatbot, and AI chatbot using a shared knowledge base. Forty participants completed information-seeking tasks and evaluated each interface. The retrieval chatbot received the highest ease-of-use rating and required the least reported effort, while the AI chatbot scored highest in perceived speed, clarity, trust, and satisfaction. Both chatbots received more favorable ratings than the FAQ webpage across most measures. These findings highlight the importance of balancing usability, effort, and user trust when designing student-information services.

Background and Purpose

Background

- Universities use FAQ webpages and chatbots to provide information on admissions, registration, fees, and student services (Peyton et al., 2025).
- FAQ webpages provide stored answers, while chatbots offer conversational approaches to information retrieval (Han & Lee, 2022; Oliveira & Matos, 2023).
- Previous studies have examined differences in usability, accuracy, speed, and mental effort across information-seeking systems (Suherwin et al., 2026; Stadler et al., 2024).
- However, no study compares FAQ webpages, retrieval chatbots, and AI chatbots using the same student-service information.

Goal

- To compare the user experience of three student-service interfaces using a shared knowledge base.

Figure 1: Search Frustration



Objectives

- Develop an FAQ webpage, retrieval chatbot, and AI chatbot using the same student-service knowledge base.
- Compare perceived speed, mental effort, ease of use, clarity, trust, accuracy, and satisfaction across the interfaces.
- Identify users' preferred interface and the reasons for their preferences.

Figure 2: Cognitive Overload



Methodology

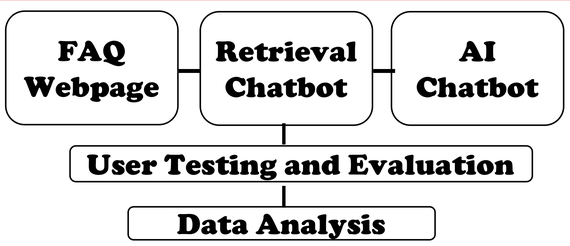
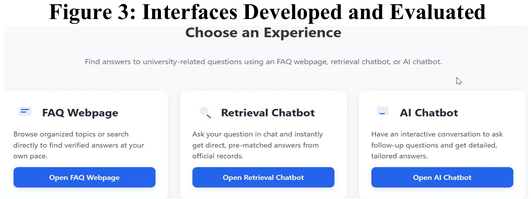


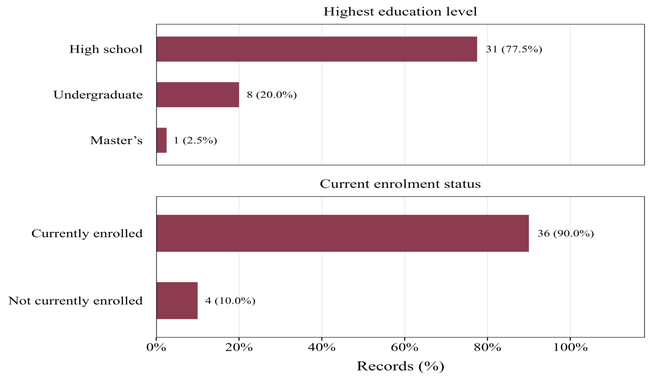
Figure 3: Interfaces Developed and Evaluated



Results and Discussion

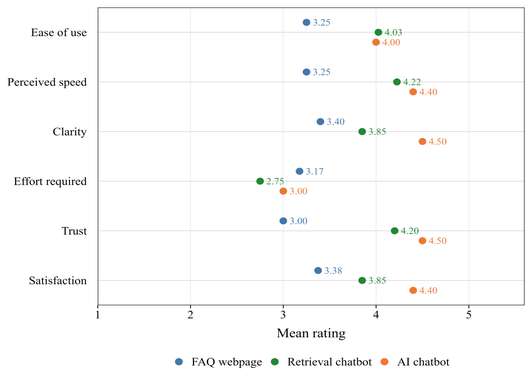
Overall Findings

Figure 4: Participants Demographics



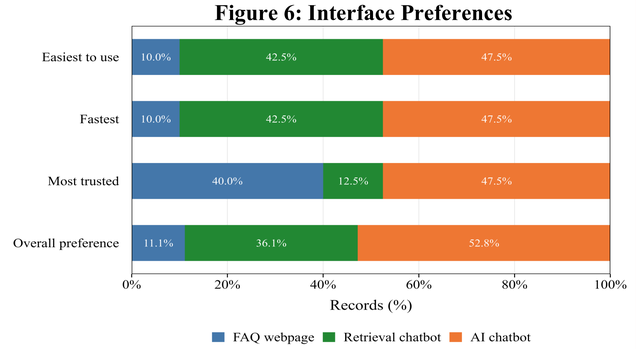
- Of the 40 participants, 90% were currently enrolled in an educational program.
- Most had completed high school (77.5%), followed by undergraduate (20%) and master's (2.5%) degrees.

Figure 5: User-Experience Ratings



- The retrieval chatbot was rated easiest to use and required the least reported effort, while the AI chatbot scored highest in perceived speed, clarity, trust, and satisfaction.
- Both chatbots received more favorable ratings than the FAQ webpage across most measures.

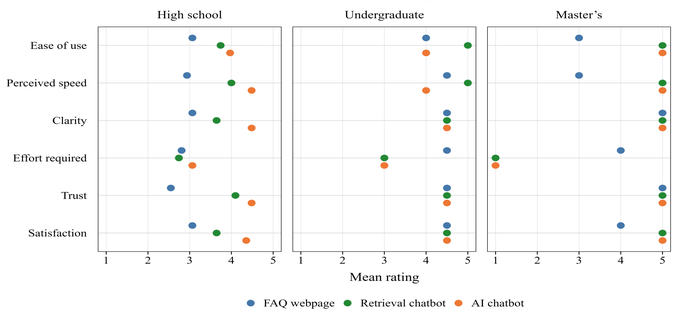
Figure 6: Interface Preferences



- The AI chatbot was most frequently selected as easiest to use, fastest, and most trusted (47.5% each).
- Overall, the AI chatbot was preferred by 52.8% of respondents expressing a preference, followed by retrieval (36.1%) and FAQ (11.1%).

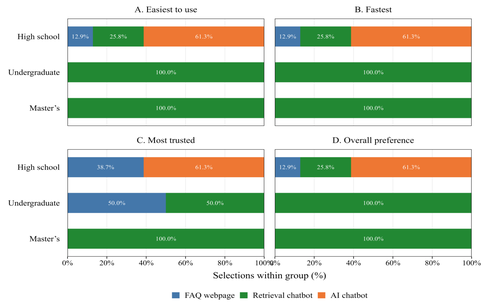
Education-based Comparison

Figure 7: User-Experience Ratings by Education Level



- AI generally received higher ratings among high-school graduates, while retrieval showed advantages in ease of use and perceived speed among undergraduates.

Figure 8: Interface Preferences by Education Level



- High-school graduates generally preferred the AI chatbot, while participants with undergraduate or master's degrees favored the retrieval chatbot.

Conclusions and Future Work

Conclusion

- Both chatbots outperformed the FAQ webpage in most user-experience ratings.
- Retrieval led in ease of use and lower reported effort, while AI led in perceived speed, clarity, trust, and satisfaction.
- No single interface excelled across all measures.

Future

- Test with real university information and a larger, more balanced sample.
- Incorporate behavioral measures to better assess user performance.

References

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